MarketBudget

Seven-day vegetable procurement budgeting for Dambulla.

The problem and the user

A small Sri Lankan caterer or canteen needs a weekly cash plan before future prices are known. The purchasing person enters planned kilograms of beans, carrots and tomatoes and an optional limit. MarketBudget provides a dated benchmark estimate and nominal 80% joint range; suppliers, quantities and purchasing decisions stay with the buyer.

The product delivered

A responsive five-view dashboard, editable basket planner, budget exposure, price explorer, saved baskets, observed outcome comparison, dated CSV/JSON exports and a visible source correction trail. The local FastAPI service stores records in SQLite; the private hosted replay saves them in the browser. Validated backups restore either edition. Ready-to-run artifacts need no paid inference API.

Real sources, visible boundaries

Recovered vegetablesSriLanka 1.1.0 data provides 2,926 valid quotes in 2,958 retained 2022+ rows through 10 March 2026. Three July 2024 corrections were verified against the CBSL wholesale Today column. Fifteen primary quotes were checked across 2023–2026. The 2022 sample audit remains incomplete. NASA POWER supplies three regional weather proxies with a seven-day availability buffer.

A precise experiment

Targets average valid observed point quotes on issue+1 through issue+7, with at least four quotes per vegetable. Missing labels remain missing. Train in 2022–2023, select in Jan–Jun 2024, calibrate in Jul–Dec 2024, test on 2025 and check early 2026. Purge labels crossing split boundaries. Three baselines compete against four restricted LightGBM quantile variants.

This is an explicit historical replay. A single October 2026 reference report is retained separately and does not imply continuous recent coverage or a live quotation.

Results worth showing

2025 BASKET MAE	NOMINAL 80% COVERAGE	EVALUATED BASKETS
LKR 2,674	78.1%	201

The release retains a baseline

The strongest development model reduced basket MAE by 16.96%, but breached the tomato error guard. The release therefore selects the latest-price baseline with a joint range from 105 aligned calibration residuals. It does not claim that adding machine learning automatically improves purchasing.

On 201 complete 2025 issue dates for the fixed 20 kg beans / 30 kg carrots / 20 kg tomatoes basket, MAE was LKR 2,674.05, signed bias LKR -52.11, observed nominal-80% coverage 78.11%, and mean interval width LKR 8,855.59. These are benchmark results, not paid invoice performance or savings.

A separate six-fold study of 20 fixed candidates found a best observed 6.60% development gain. Selected nominal-80% development coverage was 71.67%. Research promotion remains disabled. See the standalone research report for guard decisions and uncertainty.

Reliability and practical limits

Unrecognized source layouts enter review. Approved data/model bundles are immutable and hash-verified. Exact decimal calculations, transactional multi-tab saves, backup recovery and accessible chart controls are implemented. Current mode withholds old contracts or quotes older than three calendar days. Future-data invariance, target boundaries, stale inputs, saved exports and browser/API numerical parity are tested. Browser QA covers desktop and mobile views.

Upstream CBSL training and redistribution rights remain unconfirmed. Historical publication times and weather vintages are unavailable. The source audit is partial and interval evidence relies on overlapping dates. No participating buyer, prospective pilot, verified savings or national impact has been established.

Run it; then validate it prospectively

Install: python -m pip install -e '.[dev]'. Build: python scripts/build.py. Start: python -m marketbudget.api. Open http://127.0.0.1:8000. README includes reproduction instructions; DEMO_SCRIPT.md provides a three-minute walkthrough. Next milestones are rights confirmation, a continuous audited recent feed, buyer recruitment and a four-to-eight-week shadow pilot.

Credits: Talagala, Jayarathne and Talagala, vegetablesSriLanka 1.1.0; CBSL; NASA POWER; LightGBM and the Python/browser stack. New code GPL-3.0-or-later; third-party rights remain separate. OpenAI Codex assisted in implementation and documentation. Submission review and ownership remain with the original team.